Huffman Coding

Consider the fol i/p string representing a dataset label.

"DATA ANALYTICS AND INTELLIGENCE LABORATORY"

Apply huffman coding algorithm to compress this string. Also construct a huffmann tree by showing step by step merging.

Step 1:- Frequency Analysis

Letter	Count
A	7
L	4
N	4
T	4
E	3
I	3
C	2
D	2
O	2
R	2
Y	2
B	1
G	1
S	1

Step 2:- Arrange them in Ascending order

B(1) - G(1) - S(1) - C(2) - D(2) - O(2) - R(2) - Y(2) - E(3) - I(3) - L(4) - N(4), T(4), A(7).

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Step 3:- Apply huffman algorithm to build the huffman tree

Initially:

B[1]-G[1]-S[1]-C[2]-D[2]-O[2]-R[2]-Y[2]-E[2]-I[3]-L[4]-N[4]-T[4]-A[7].

Itr1:- Combine B[1] and G[1] to get BG[2]

S[1]-C[2]-D[2]-O[2]-R[2]-Y[2]-BG[2]-E[3]-I[3]-L[4]-N[4]-T[4]-A[7]

Itr2:- Combine S[1] and C[2] to get SC[3]

D[2]-O[2]-R[2]-Y[2]-BG[2]-E[3]-I[3]-SC[3]-L[4]-N[4]-T[4]-A[7]

Itr3:- Combine D[2] and O[2] to get DO[4].

R[2]-Y[2]-BG[2]-E[3]-I[3]-SC[3]-L[4]-N[4]-T[4]-DO[4]-A[7].

Itr4:- Combine R[2] and Y[2] to get RY[4].

BG[2]-E[3]-I[3]-SC[3]-L[4]-N[4]-T[4]-DO[4]-RY[4]-A[7].

Itr5:- Combine BG[2] and E[3] to get BGE[5]

I[3]-SC[3]-L[4]-N[4]-T[4]-DO[4]-RY[4]-BGE[5]-A[7].

Itr6:- Combine I[3] and SC[3] to get ISC[6].

L[4]-N[4]-T[4]-DO[4]-RY[4]-BGE[5]-ISC[6]-A[7].

Itr7:- Combine L[4] and N[4] to get LN[8].

T[4]-DO[4]-RY[4]-BGE[5]-ISC[6]-A[7]-LN[8]

Itr8:- Combine T[4] and DO[4] to get TDO[8]

RY[4]-BGE[5]-ISC[6]-A[7]-LN[8]-TDO[8].

Itr9:- Combine RY[4] and BGE[5] to get RYBGE[9].

ISC[6]-A[7]-LN[8]-TDO[8]-RYBGE[9].

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Itu 10:- Combine ISC[6] and A[7] to get ISCA[13].

LN[8]-TDO[8]-RYBGE(9]-ISCA[13].

Itu 11:- Combine LN[8] and TDO[8] to get LNTDO[16].

RYBGE[9]-ISCA[13]-LNTDO[10]

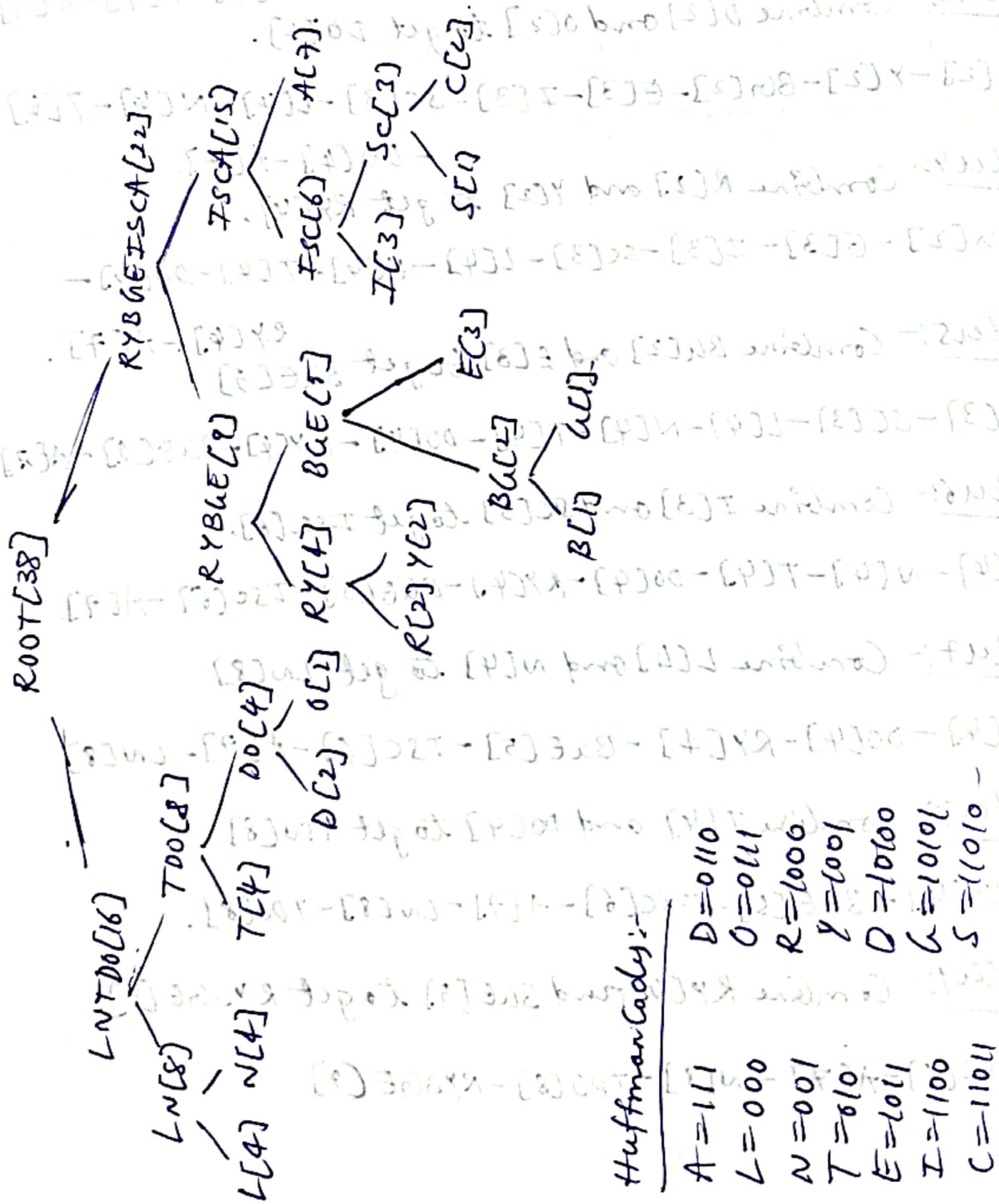
Itu 12:- Combine RYBGE[9] and ISCA[13] to get

RYBGEISCA[22].

LNTDO[16]-RYBGEISCA[22].

Combine LNTDO[16] and RYBGEISCA[22] to get

LNTDORYBGEISCA[38].



Huffman Code:-

A=111
L=000
N=001
T=010
E=1011
I=1100
C=11011
D=0110
O=0111
R=1000
Y=1001
B=10100
G=10101
S=11010